

Problem 10.14

The force of interest δ is constant. The corresponding nominal rate of discount when interest is compounded p thly is $d^{(p)}$. Express $d^{(p)}$ in terms of p . Show that $d^{(p)}$ is an increasing function of p .

Solution

Under a constant force of interest, the annual accumulation factor is

$$e^{\delta}.$$

For a nominal discount rate $d^{(p)}$ convertible p times per year, the accumulation factor over one year is

$$\left(1 - \frac{d^{(p)}}{p}\right)^{-p}.$$

Equating the two accumulation factors gives

$$\left(1 - \frac{d^{(p)}}{p}\right)^{-p} = e^{\delta}.$$

Taking both sides to the power $-1/p$,

$$1 - \frac{d^{(p)}}{p} = e^{-\delta/p}.$$

Therefore,

$$\boxed{d^{(p)} = p \left(1 - e^{-\delta/p}\right)}.$$

To study monotonicity, extend this expression to positive real values of p and define

$$g(p) = p \left(1 - e^{-\delta/p}\right).$$

Differentiating,

$$g'(p) = 1 - e^{-\delta/p} \left(1 + \frac{\delta}{p}\right).$$

For $x > 0$, the inequality

$$e^x > 1 + x$$

holds. Taking $x = \delta/p$ gives

$$e^{\delta/p} > 1 + \frac{\delta}{p}.$$

Multiplying by $e^{-\delta/p}$,

$$1 > e^{-\delta/p} \left(1 + \frac{\delta}{p}\right).$$

Hence,

$$g'(p) > 0.$$

Therefore, $d^{(p)}$ is an increasing function of p .

Also,

$$\lim_{p \rightarrow \infty} d^{(p)} = \lim_{p \rightarrow \infty} p \left(1 - e^{-\delta/p} \right) = \delta,$$

so the nominal discount rates increase toward the force of interest.